

Euclid's Elements — Book I

Proposition 3

Formal Reconstruction — Technical Documentation

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Cutting off a prescribed length from a longer segment

1 Introduction

Proposition I.3 asks for a segment-copying construction with an order constraint. Given a greater segment AB and a shorter segment CD , one must find an interior point E of AB such that

$$AE \cong CD.$$

Classically, Euclid solves the problem by combining Proposition I.2 with a circle centered at A . The present formal reconstruction has a very different architecture. The hypothesis is not a primitive numerical or order-theoretic inequality. It is the witness-based predicate `HilbertSegmentLess`. Its definition already asserts the existence of a point inside the greater segment cutting off a congruent initial part.

Thus I.3 is a particularly clean example of a change of mathematical language. Euclid proves the existence of the desired point by a geometric construction; Book Zero defines strict segment comparison by the existence of exactly such a point. At the public Book I level, the Lean theorem only unpacks the witness and reverses the orientation of segment congruence.

2 Classical Configuration

Let AB be the greater segment and CD the shorter one. Euclid first uses Proposition I.2 to construct from A an auxiliary segment AF satisfying

$$AF \cong CD.$$

A circle centered at A through F then meets the segment AB at E . Since AE and AF are radii of the same circle,

$$AE \cong AF,$$

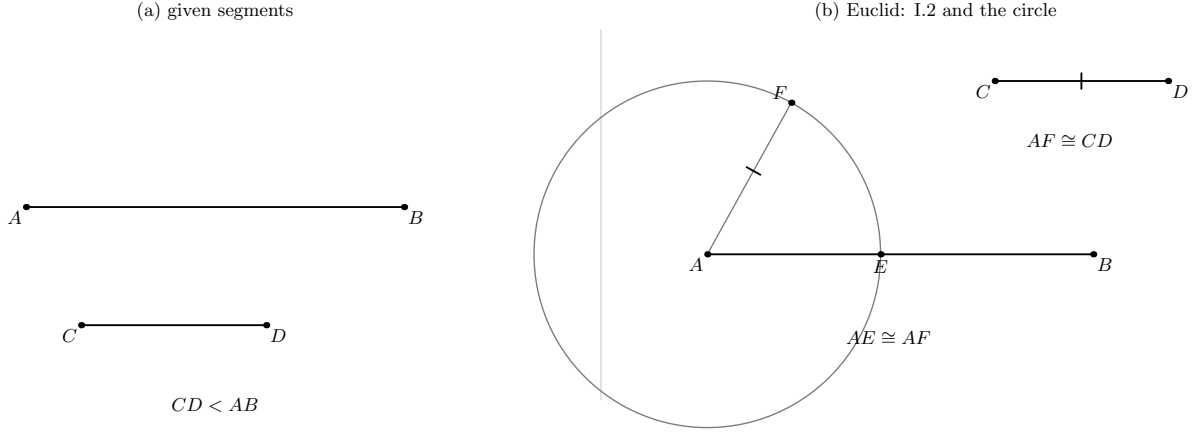


Figure 1: Euclid's construction for Proposition I.3. (a) The input consists of the greater segment AB and the shorter segment CD . (b) Proposition I.2 supplies $AF \cong CD$; the circle centered at A through F cuts AB at E , so $AE \cong AF \cong CD$.

and hence

$$AE \cong CD.$$

The inequality $CD < AB$ is what places the circle intersection on the open segment from A to B , rather than beyond B . The final positional statement is therefore

$$A - E - B.$$

3 Statement

Theorem 1 (Euclid I.3). *Let AB and CD be segments, and suppose CD is strictly shorter than AB . Then there exists a point E such that*

$$A - E - B$$

and

$$AE \cong CD.$$

The current production declaration is:

```
theorem euclid_proposition_3
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point)
  (hCDAB : HilbertSegmentLess Geo C D A B) :
  ∃ E : Geo.Point,
    Geo.Between A E B ∧
    Geo.Congruent A E C D := by

rcases hCDAB with ⟨E, hAEB, hCDAE⟩

exact ⟨E, hAEB,
  hilbert_congruent_symmetry Geo C D A E hCDAE⟩
```

Diagrammatic audit. The preserved chapter displayed the final segment configuration and a logical flowchart. The flowchart is unnecessary once the definition of `HilbertSegmentLess` is inspected, while the final configuration alone does not show Euclid’s actual construction.

The revised chapter therefore uses two geometric figures with three panels in total:

$$2 \text{ classical panels} + 1 \text{ formal panel} = 3 \text{ panels in 2 figures.}$$

The formal panel is intentionally simple: there is no hidden construction below the predicate at this point, because the predicate itself is defined by the required geometric witness.

4 Classical Proof

Start with the two unequal segments, with AB the greater. By Proposition I.2, construct from the point A a segment AF congruent to the given shorter segment CD :

$$AF \cong CD.$$

By Postulate 3, draw the circle with center A and radius AF . Let this circle meet the greater segment AB at E . Because E and F lie on the same circle centered at A ,

$$AE \cong AF.$$

Together with $AF \cong CD$, Common Notion 1 gives

$$AE \cong CD.$$

Since the radius represented by CD is shorter than AB , the intersection point lies between A and B . Hence

$$A - E - B.$$

This is the required construction.

The direct classical dependency is therefore

$$\text{I.2} \quad + \quad \text{Postulate 3 (circle)} \quad + \quad \text{Common Notion 1.}$$

Proposition I.1 is not a direct dependency of I.3.

5 Formal Meaning of Segment Inequality

The essential formal fact is the definition

```
def HilbertSegmentLess
  (Geo : Geometry.Geo)
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point) : Prop :=
  ∃ P : Geo.Point,
    Geo.Between C P D ∧
    Geo.Congruent A B C P
```

Thus

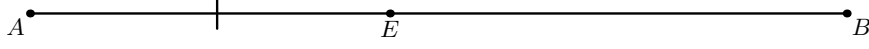
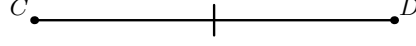
`HilbertSegmentLess Geo C D A B`

expands to

$$\exists E, \quad A - E - B, \quad CD \cong AE.$$

This is already the full existential content of Proposition I.3, except that the congruence is oriented as $CD \cong AE$ rather than $AE \cong CD$.

$$CD \cong AE$$



$$A - E - B$$

Figure 2: The geometric witness contained in `HilbertSegmentLess Geo C D A B`. The predicate already supplies $A - E - B$ and $CD \cong AE$; the public theorem only reverses the congruence orientation.

This point is foundationally important. In the formal language used here, “ $CD < AB$ ” is not merely an abstract comparison from which the cut-off point must later be constructed. The comparison is defined synthetically by the existence of an initial subsegment of AB congruent to CD .

6 Formal Proof Architecture

The complete formal chain is

```

hCDAB : HilbertSegmentLess Geo C D A B
      ↓
    unfold/extract the witness
      ↓
    E,  A - E - B,  CD ≅ AE
      ↓
    hilbert_congruent_symmetry
      ↓
    A - E - B,  AE ≅ CD
      ↓
    euclid_proposition_3.

```

There is no theorem call corresponding to Euclid’s use of I.2, and there is no formal circle construction in the proof. Nor is there a deeper hidden proof engine beneath `HilbertSegmentLess`:

it is a definition, not a theorem whose proof could conceal further geometry.

The only nontrivial operation after witness extraction is symmetry of segment congruence:

```
hilbert_congruent_symmetry Geo C D A E hCDAE
```

which changes

$$CD \cong AE$$

to

$$AE \cong CD.$$

7 Relationship with Earlier Propositions

7.1 Proposition I.2

Proposition I.2 is the decisive earlier proposition in Euclid’s proof. It allows the length of the shorter segment CD to be reproduced from the specified point A , producing the auxiliary radius AF .

The formal proof of I.3 does not call `euclid_proposition_2`. The constructional content that I.2 supplies classically has already been absorbed into the witness-based language of strict segment comparison.

7.2 Proposition I.1

I.1 is not a direct dependency of Euclid’s I.3. Any appearance of the equilateral-triangle construction occurs inside Euclid’s proof of I.2, not as a new step of I.3 itself. The classical dependency should therefore be recorded as

$$I.2 \longrightarrow I.3,$$

not as a direct edge from I.1 to I.3.

7.3 Transition to Proposition I.4

I.3 closes the initial sequence of explicit segment constructions. I.4 then changes character completely: it is a rigidity theorem for triangles rather than a point-construction theorem. In the formal development the contrast is even sharper, because I.3 is witness extraction from a definition, while I.4 is a wrapper around the developed Hilbert SAS theorem.

8 Hilbert and Book Zero Components Used

8.1 HilbertSegmentLess

This definition is the central object of the formal proposition:

$$AB < CD \iff \exists P (C - P - D \wedge AB \cong CP).$$

It provides a synthetic notion of strict segment comparison without introducing numerical lengths.

For I.3 the arguments are instantiated in the reversed role

$$CD < AB,$$

so the witness lies inside AB and cuts off a segment congruent to CD .

8.2 Segment Congruence Symmetry

The definition returns

$$CD \cong AE,$$

while the proposition states

$$AE \cong CD.$$

The theorem `hilbert_congruent_symmetry` is therefore the only reusable theorem called after the witness is unpacked.

8.3 Betweenness

The open-segment condition is represented directly by `Geo.Between`. Consequently the phrase “cut off from the greater straight line” is not inferred from a picture: the output explicitly contains

$$\text{Geo.Between } A \ E \ B.$$

9 Formalization Notes

Proposition I.3 is one of the clearest examples in Book I where theorem size can be misleading. The Lean proof is extremely short, but not because a circle construction has been compressed into automation. Rather, the input predicate has been chosen so that its witness is exactly the output point required by the proposition.

This is a legitimate and useful synthetic design choice. The relation `HilbertSegmentLess` is not numerical: it defines strict comparison by internal geometric configuration. It therefore supports later proofs by turning inequalities immediately into betweenness and congruence data.

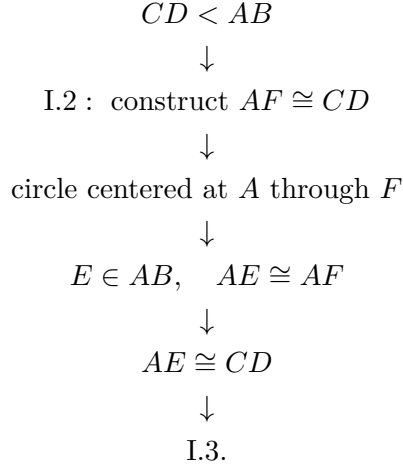
At the same time, documentation must keep the two layers distinct:

- Euclid constructs the witness using I.2 and a circle;
- Book Zero uses the existence of such a witness as the definition of strict segment comparison;
- I.3 then exposes that witness as a Book I theorem.

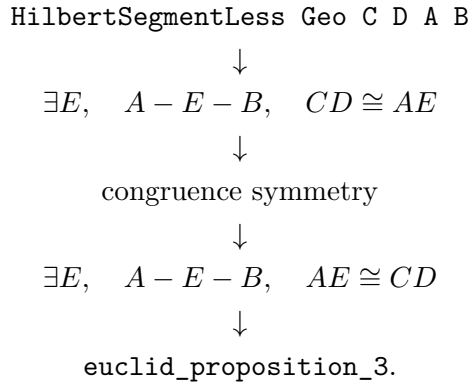
The diagrammatic audit reflects this distinction. The classical figure shows the construction, while the formal figure shows only the geometric state encoded by the comparison predicate. A logical flowchart would add no new geometric information.

10 Dependency Summary

The classical architecture is



The formal architecture is much shorter:



No local axiom is introduced by `euclid_proposition_3`; the theorem uses the Book Zero definition of strict segment comparison and the existing congruence-symmetry theorem.

11 Conclusion

Proposition I.3 completes Euclid’s first sequence of segment constructions. Classically, it reuses I.2 and adds one circle to cut a prescribed length from a greater segment. Formally, the same geometric endpoint has already been built into the definition of strict segment comparison.

The proposition therefore illustrates a recurring theme of the reconstruction: a classical construction may reappear not as a long theorem proof but as the witness structure of a carefully chosen synthetic predicate. The short Lean proof is mathematically informative precisely because the underlying notion `HilbertSegmentLess` has made the relevant geometry explicit in its definition.